

MBI Signal Logger Operational Manual

Public, Field and Admin operating workflows

Portfolio context

Developed during my IT internship at Murphy Ben International (MBI) as an internship/internal experimental project for collecting and analyzing broadcast signal reports. I was responsible for the design and implementation of the application, working under the guidance of a senior member of the IT team.

This repository is a portfolio representation of the project. Sensitive configuration, internal information, production data, credentials, and company-specific deployment details have been excluded.

1. System overview

MBI Broadcast Signal Logger collects structured broadcast reception reports from radio listeners and field personnel, then makes the combined information available for administrative review, mapping and export.

Application	Route	Primary use
Public Logger	/	Quick listener and fan reports
Field Logger	/field/	GPS-assisted engineering observations
Admin Center	/admin/	Operations, analytics, configuration and exports

Operational flow

Observation -> validation -> PUB/INC identity -> GPS/RF enrichment -> storage -> Admin review

2. Public reporting workflow

- Open the Public Logger and dismiss the introduction.
- Allow location access so the report includes current GPS context.
- Select the station and optional channel or frequency.
- Choose Signal Quality, Signal Stability and Service Condition.
- Add an optional observation, review the summary and submit.
- Record the returned PUB incident ID when follow-up is required.

3. Field reporting workflow

- Open New Incident in the Field Logger.
- Confirm the engineer or operator identity used for the report.
- Capture GPS and verify the displayed accuracy.
- Select the station, channel and reception observations.
- Add measured values or custom fields when available.
- Set priority and assignment where configured, then submit.
- Use history, mapping and offline controls as needed.

4. Admin operations

Area	Operational purpose
Dashboard	Totals, trends, categories and recent incidents
Incidents	Search, filter, inspect, assign and update reports
Stations and channels	Shared reference information for both reporting apps
Maps	Compare report coordinates and configured stations
Exports	CSV, print and Google Earth KML outputs
App configuration	Public and Field presentation, forms and workflow
Health and audit	Storage status and operational activity

5. Report interpretation

Public and Field reports remain separate at storage level but are available together through combined analysis. Observation scores provide a comparable reception index. RF values are analytical references and are not guaranteed coverage predictions.

6. Health and troubleshooting

```
GET https://signal-logger.example.com/api/health
```

- Confirm IIS returns the frontend pages over HTTPS.
- Confirm the health endpoint reports healthy runtime stores.
- Verify new PUB and INC reports persist and appear in Admin.
- Use Ctrl+Shift+R or Ctrl+F5 after frontend changes.
- If Field remains stale, clear site data before considering service-worker removal.

7. Operating limits

Protect Admin access in a real deployment. Keep credentials, certificates, production data, station coordinates and private infrastructure outside the public repository.